

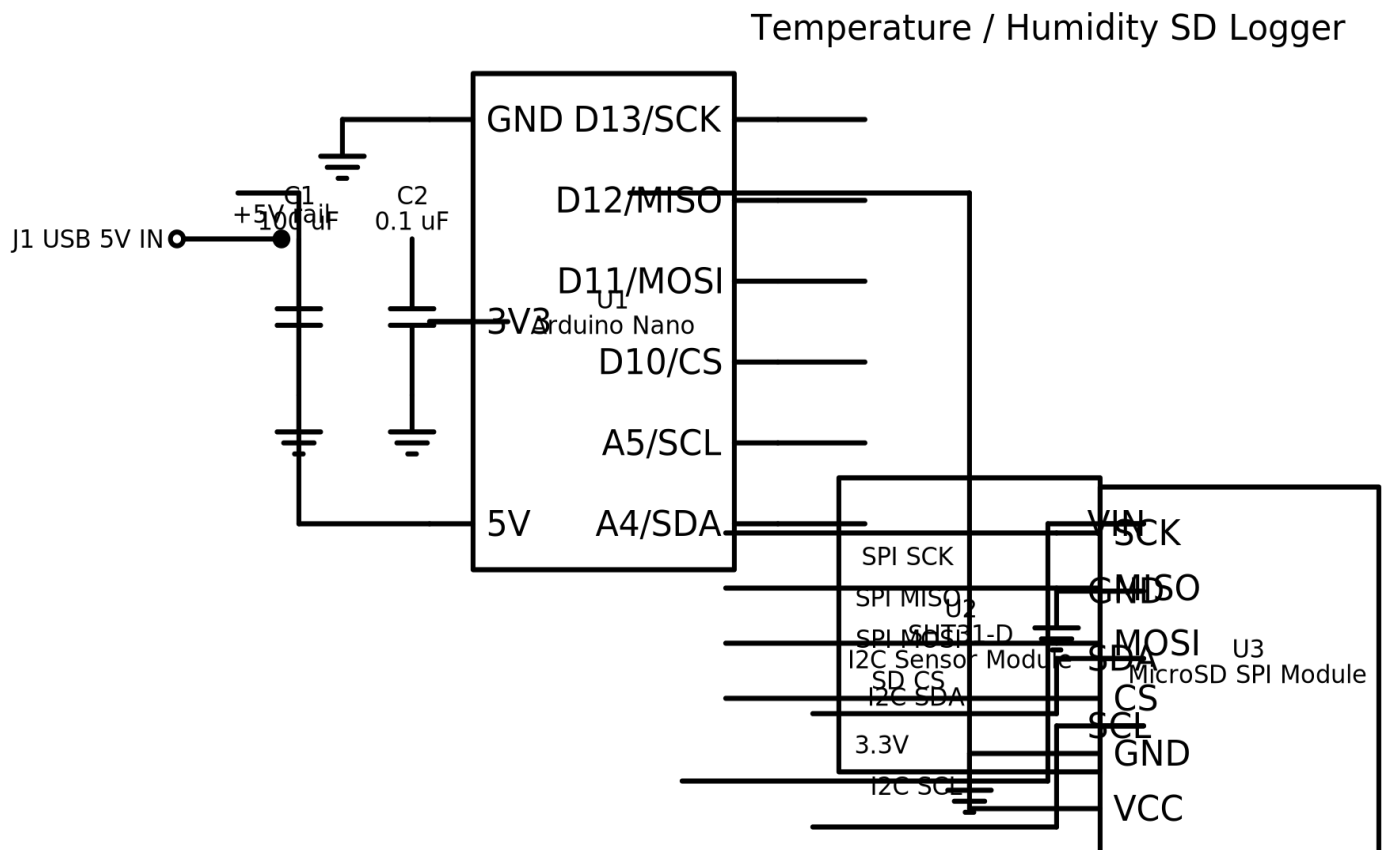
Temperature and Humidity Logger with microSD Storage

USB-powered Arduino Nano logger using an SHT31-D digital sensor and SPI microSD module.

Assumptions

- USB-powered 5 V design intended for indoor or sheltered use.
- Cost was prioritized over premium metrology performance while still targeting practical accuracy near 0.1 °C display resolution and about 1-2 %RH class sensor performance.
- Sensor chosen: Sensirion SHT31-D breakout/module because it is typically more accurate and more stable than DHT-class parts while remaining affordable.
- Microcontroller chosen: Arduino Nano compatible board for easy SD logging and broad library support.
- MicroSD module assumed to be a common SPI breakout with onboard 3.3 V regulator and level shifting for 5 V Arduino compatibility.
- This is a conceptual but buildable package, not a certified production release.

Schematic



All grounds common

Firmware files

Arduino sketch: temp_humidity_sd_logger.ino

The sketch logs elapsed milliseconds, temperature in °C, and humidity in %RH once per minute with 0.1-unit formatted output.

Bill of Materials

RefDes	Description	Qty	Unit \$	Ext. \$	Supplier / Link
U1	Arduino Nano compatible board (ATmega328P, 5 V)	1	6.50	6.50	Amazon/Elegoo Nano compatible https://www.amazon.com/s?k=arduino+nano+compatible
U2	SHT31-D temperature/humidity breakout, I2C, 3.3-5 V tolerant module	1	7.50	7.50	Adafruit/SparkFun-compatible module https://www.adafruit.com/product/2857
U3	MicroSD card module, SPI, 5 V interface	1	2.50	2.50	Catalex-style microSD module https://www.amazon.com/s?k=micro+sd+card+module+arduino
J1	USB Mini/Micro cable or Nano USB cable	1	2.00	2.00	Generic https://www.amazon.com/s?k=usb+cable+for+arduino+nano
C1	100 uF electrolytic capacitor, >=10 V	1	0.20	0.20	Generic https://www.digikey.com/en/products/filter/aluminum-electrolytic-capacitors/58
C2	0.1 uF ceramic capacitor	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/ceramic-capacitors/60
SD1	MicroSD card, 8-32 GB	1	4.00	4.00	SanDisk or equivalent https://www.amazon.com/s?k=micro+sd+card+16gb
HW1	Breadboard or perfboard + jumper wires/header pins	1	3.00	3.00	Generic https://www.amazon.com/s?k=breadboard+jumper+wires
	Total estimated cost			25.75	

Limitations / notes

Live supplier pricing and stock were not verified; listed prices are reasonable estimates for single-quantity hobbyist purchases.

Absolute sensor accuracy depends on the exact module vendor, airflow, placement, and calibration. The firmware writes values with 0.1-unit resolution, which is not the same as guaranteed absolute accuracy.

For long-term deployments, add a real-time clock and enclosure design review.